Certified Defender Move Restriction in Hexo Threat Search

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Abstract

Threat-space search in Hexo has large defender branching because every legal reply must ordinarily be considered. We develop a horizon-parameterized certificate that restricts defender replies while preserving a proved attacker win over the full legal move set. Companion results establish the impossibility of a global pairing defense, a potential-function layer for blocking certificates, local domination patterns, and mechanized validation of the finite claims and move-set closure.

1 Introduction

A df-pn threat-space solver must, at a defender AND node, in principle consider every legal reply. In Hexo this means every empty cell within hex distance 8 of any existing stone, often hundreds of cells. Enumerating these replies dominates the cost of the search.

The restriction studied here is *certified*: every omitted defender reply has a verifier-checkable proof of sound dismissal. Search trimming may cause the solver to miss an attacker win, so it can reduce completeness. It cannot cause a false attacker win to pass verification. This separation between completeness and soundness holds regardless of the heuristic used to generate the searched replies.

The restriction must depend on a finite horizon. No fixed static set is sufficient for every future continuation on the unbounded board. A certificate instead states its own placement horizon and carries the zone obligations needed up to that horizon.

The paper first gives the formal game model in Section 2 and the locality lemmas in Section 3. It then defines zone-carrying certificates in Section 4 and proves horizon-parameterized dismissal soundness in Section 5. The computable constructions of Section 6 include a measured generator mean of 147 cells, compared with 302 legal cells. The remaining layers establish the impossibility of a global pairing defense in Section 7, develop potential-function certificates in Section 8, give local domination patterns in Section 9, and report mechanized validation in Section 10.

Threat-space search and proof-number methods provide the search setting for these certificates [1, 2]. Pairing strategies and the potential method are standard tools for positional games [3, 4].

The full internal proof documents, review history, and validation scripts are available in the repository [5].

2 The game and formal model

2.1 Informal rules

Hexo is a two-player placement game on an unbounded hexagonal grid. The opening player places one stone at the origin. Thereafter the players alternate turns of two placements. Stones are permanent. A placement is legal on any empty cell within hex distance 8 of a stone of either player. The first player to occupy six consecutive cells on one of the three grid axes wins. A win is checked after each placement, so a win on the first placement ends the game before the second placement of that turn.

We use attacker and defender as fixed roles, denoted by A and D. A ply is one placement, while a normal turn contains two plies. Everything below concerns positions after the Opening placement (the origin is occupied; budgets follow the turn structure of Definition 2.5).

2.2 Formal model

The first definition fixes the infinite board and the metric used by both the line geometry and the legality rule.

Definition 2.1 (Board, cells, and distance)**.** The cells are the integer axial lattice $\mathbb{Z}^2 \ni x = (q, r)$. For $x = (q, r)$ and $y = (q', r')$, set

$$d(x, y) = \max\{|q - q'|,\ |r - r'|, |(q - q') + (r - r')|\}.$$

The winning patterns are finite segments along the three line directions of the axial lattice.

Definition 2.2 (Axes and windows)**.** The axis vectors are

$$\mathfrak{a} \in \{(1, 0), (0, 1), (1, -1)\}.$$

A *window* is a set

$$W = \{s + i\mathfrak{a} : 0 \le i \le 5\}$$

for a start cell s and an axis vector $\mathfrak{a}$. Every window has exactly six cells, and every cell lies in exactly 18 windows, from three axes and six possible offsets on each axis.

The next fact records the metric consequence of belonging to one window.

Fact 2.3 (Window collinearity)**.** Two distinct cells of a window differ by a nonzero multiple of the window's axis vector. Thus any two cells contained in a common window are collinear along that axis and have distance at most 5.

Figure 1 shows the coordinate convention and one distance calculation.

A position records the finite board state and the exact placement remaining in the current turn.

Definition 2.4 (Positions)**.** A position is $P = (\sigma, m, b)$, where σ is a finite partial map $\mathbb{Z}^2 \rightharpoonup \{\mathrm{A}, \mathrm{D}\}$, the mover is $m \in \{\mathrm{A}, \mathrm{D}\}$, and $b \in \{1, 2\}$ is the mover's number of remaining placements in the current turn. Stones are permanent, and $\mathrm{Stones}(P) = \operatorname{dom} \sigma$. For $X \in \{\mathrm{A}, \mathrm{D}\}$, let $\bar{X}$ denote the other player. For such an X and a window W, define

$$\mathrm{cnt}_X(W, P) = |\{c \in W : \sigma(c) = X\}|.$$

When the position is clear from context, we suppress it and write $\mathrm{cnt}_X(W)$.

Legality is local to the existing stones, while a transition must also account for immediate termination and the two-placement turn structure.

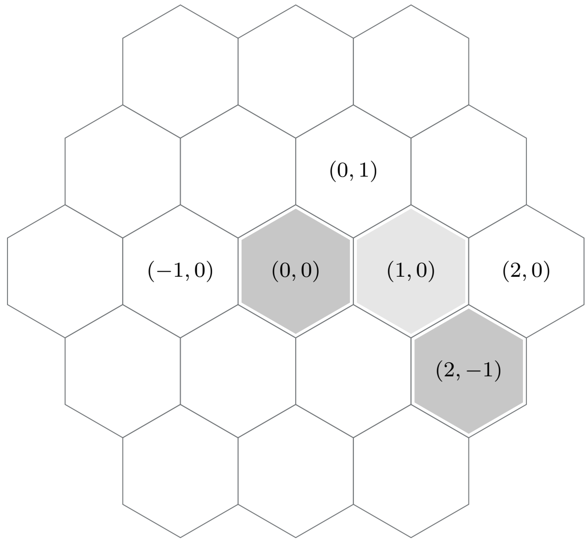
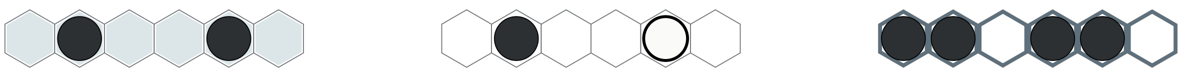


Figure 1: Pointy-top axial coordinates. The shaded cells show a shortest path from $(0, 0)$ to $(2, -1)$, and $d = \max(|2|, |-1|, |1|) = 2$.



(a) Alive for A.

(b) Dead.

(c) An A-threat with count 4.

Figure 2: Alive, dead, and threat window states. Filled disks are attacker stones and open disks are defender stones.

Definition 2.5 (Legality and transitions)**.** A cell c is *legal* in P if and only if

$$c \notin \operatorname{dom} \sigma \quad \text{and} \quad \exists s \in \operatorname{dom} \sigma : d(c, s) \leq 8.$$

Write $\operatorname{Legal}(P)$ for the set of cells legal in P. The mover places on a legal cell c, producing $\sigma' = \sigma \cup \{c \mapsto m\}$. If some window W then satisfies $\operatorname{cnt}_m(W) = 6$, the game terminates immediately and m wins. Wins are therefore checked after each placement and may occur on the first placement of a two-placement turn. If there is no win, the next position is $(\sigma', m, 1)$ when $b = 2$, and $(\sigma', \bar{m}, 2)$ when $b = 1$.

Window states distinguish usable winning lines, blocked lines, and lines that can be completed within one normal turn.

Definition 2.6 (Window states)**.** A window W is *alive for* X if $\operatorname{cnt}_{\bar{X}}(W) = 0$. An all-empty window is alive for both players. It is *dead* if $\operatorname{cnt}_{\mathrm{A}}(W) \geq 1$ and $\operatorname{cnt}_{\mathrm{D}}(W) \geq 1$. It is an X*-threat* if it is alive for X and $\operatorname{cnt}_X(W) \geq 4$.

The three panels in Fig. 2 give the basic mask patterns used by this terminology.

The implemented one-ply analysis tests whether the mover can win in the remaining placements or whether the opponent's threats exceed the defensive budget.

Definition 2.7 (λ^1 predicates)**.** For mover m with budget b, own_win_now(P) holds if and only if there is an m-alive window W such that

$$\operatorname{cnt}_m(W) = 5, \qquad \text{or} \qquad \operatorname{cnt}_m(W) = 4 \ \text{ and } \ b = 2.$$

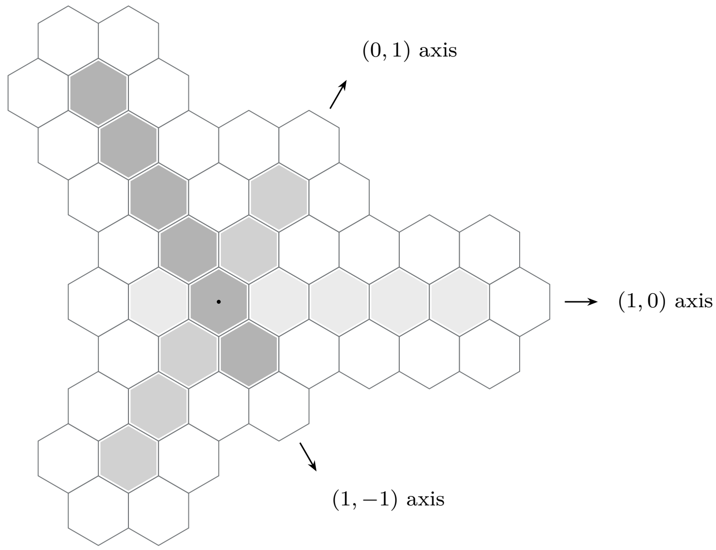


Figure 3: One six-cell window on each axis through the central cell, shown with three fill shades. Sliding a window so that the cell occupies offsets 0 through 5 gives 3 axes $\times$ 6 offsets = 18 windows through the marked cell.

The value hit(P) is the multiset of empty-cell sets of the $\bar{m}$-threat windows. The value mhs(P) is the minimum number of cells meeting every set in hit(P), computed exhaustively for values at most 2. The λ^1 verdict is WIN when own_win_now(P) holds, LOSS when it does not hold and mhs(P) $> b$, and undecided otherwise. The post-opening soundness of these implemented verdicts is assumed here.

Absolute placement indices allow later certificates to state an exact win horizon without treating a two-placement turn as simultaneous.

Definition 2.8 (Plies and horizon)**.** A *ply* is one placement, and ply indices are absolute along a play. For a position P with mover m, the *defender-placement count before ply* T, written $\mathfrak{D}(P, T)$, is the number of plies strictly before T at which D places, as determined by (m, b) and alternation.

The final convention fixes the direction needed when defender replies are removed from an attacker-win certificate.

Definition 2.9 (Dominance direction)**.** Outcomes are evaluated for the attacker A, with WIN $\succ$ nonWIN. A *sound dismissal* of a defender reply c at a node N means that, if the certificate proves that A wins against all searched replies, then A also wins against c.

3 Elementary geometry

The first bound relates the number of stones in a window to the maximum gap from an empty cell to those stones.

Lemma 3.1 (Packing bound)**.** *In a window with $k \geq 1$ stones of one colour and no stones of the other, every empty cell of the window is within distance $6 - k$ of a same-window stone. The bound is tight.*

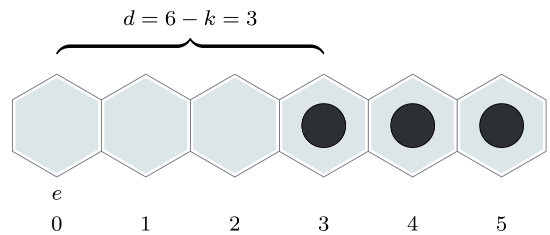


Figure 4: The tightness configuration for the packing bound, shown at $k = 3$. In general the k stones occupy offsets $6 - k$ through 5, while the empty cell at offset 0 is at distance $6 - k$ from the nearest stone.

Proof. Index the cells $0, \ldots, 5$ along the window axis. Let e be empty, and let δ be its axis-distance from a nearest in-window stone. The δ cells from e toward that stone, including e and excluding the stone, are empty and lie in the window. There are $6 - k$ empty cells in total, so $\delta \leq 6 - k$. By fact 2.3, axis-distance agrees with hex distance for these cells. For tightness, put the stones at offsets $6 - k, \ldots, 5$ and take e at offset 0. □

Figure 4 displays the extremal arrangement and a concrete $k = 3$ instance of the general formula.

A mixed set of occupied and empty cells on a finite line segment must change state across some adjacent pair.

Lemma 3.2 (Boundary pair)**.** *Any window containing at least one stone and at least one empty cell has an empty cell at distance* 1 *from a same-window stone.*

Proof. The six cells are consecutive on a line. The stone cells form a nonempty proper subset of this sequence. At a boundary of that subset, a stone and an empty cell are consecutive. Their hex distance is 1 by fact 2.3. □

The packing bound places all tactics that can resolve within the current turn near stones already on the board.

Theorem 3.3 (Single-turn tactical locality)**.** *Every placement that completes a win within the current turn, and every empty cell of every threat window, lies within distance* 2 *of an existing stone.*

Proof. Completion within at most two placements requires a window with at least four stones of the mover already present. Apply Lemma 3.1 with $k = 4$ to its empty cells. A threat window also has at least four stones by Definition 2.6, so the same bound applies to every one of its empty cells. □

Creating a threat can start one cell farther away because the placement need only raise an existing three-stone window to count four.

Theorem 3.4 (Threat-creation locality)**.** *A placement creating an X-threat lies within distance* 3 *of an X stone. The bound is tight.*

Proof. Immediately before the placement, the window that becomes an X-threat held at least three X stones. Apply Lemma 3.1 with $k = 3$ to the placement cell. For tightness, place X stones at offsets $3, 4, 5$ of a window and make the threat-creating placement at offset 0. □

4 Zone-carrying certificates

At an AND node, the solver may face hundreds of legal defender replies. We want a searched set at each such node for which every omitted reply carries a proof of dismissal. No unbounded static set can provide this guarantee, so the construction is parameterized by the certificate's own placement horizon.

4.1 Transfer lemmas

Every comparison of two positions must account for the same three ways in which a stone can affect the game.

Lemma 4.1 (Difference accounting)**.** *Let $P' = P \pm \{x\}$ differ from P by one stone at the cell x. Every game-relevant predicate difference between P and P' is confined to the following channels:*

(C1) the occupancy of x, including its availability as a move and its ownership;

(C2) the masks of the 18 windows containing x, and hence every window-mask-derived predicate: completion, alive or dead status, threat status, own_win_now, hit, and mhs;

(C3) the legality frontier

$$\{c : d(c, x) \leq 8\} \setminus \operatorname{dom} \sigma,$$

whose cells may gain or lose a legality justification through x.

Proof. By Definition 2.5, the transition relation depends on occupancy at the placed cell, the legality predicate, and the window masks used for termination. By Definitions 2.6 and 2.7, every tactical predicate is a function of window masks. Only windows containing x change masks. By Definition 2.5, only cells within distance 8 of x can gain or lose a legality justification through x. There are no other state components. □

Thus a transfer argument is complete exactly when it accounts for occupancy, incident-window masks, and the legality frontier.

Permanence lets later witnesses be interpreted at earlier points of the same play.

Lemma 4.2 (Permanence)**.** *Window masks are monotone nondecreasing along every play. Dead windows remain dead. A window alive for X at ply t is alive for X at every earlier ply.*

Proof. Stones are never removed by Definitions 2.4 and 2.5. □

Adding inert defender stones transfers the cells, masks, legality, and completion events used later in the proof.

Lemma 4.3 (Inert-extension transfer)**.** *Let $P' = P + X$, where X is a finite set of defender stones. Let $\mathcal{W}$ be a set of windows and C a set of cells such that*

(i) no $x \in X$ lies in any $W \in \mathcal{W}$; and

(ii) $X \cap C = \varnothing$.

Then:

(a) every cell of C empty in P is empty in P';

(b) every $W \in \mathcal{W}$ has identical masks in P and P';

(c) $\mathrm{Legal}(P) \setminus X \subseteq \mathrm{Legal}(P')$; in particular, every cell of C legal in P is legal in P'; and

(d) the A-complete windows of P' are exactly the A-complete windows of P that are disjoint from X.

Consequently, an A*-completion on a window known to be disjoint from X transfers from P to P', and every* A*-completion in P' was available in P.*

Proof. Parts (a) and (b) follow from occupancy and incident-window masks in Lemma 4.1. For (c), adding stones can only add legality justifications, while occupancy removes precisely the cells of X. For (d), an A-complete window must contain no defender stone. Windows meeting X are blocked in P', and the masks of the other windows agree. □

Deleting defender stones is monotone for attacker tactics, but it requires a separate legality condition for attacker placements.

Lemma 4.4 (Deletion monotonicity and well-formed legality)**.** *Let $P'' = P - Y$, where Y is a finite set of defender stones of P. Then:*

(a) every A-threat window of P is an A-threat window of P'', so $\mathrm{mhs}(P'') \geq \mathrm{mhs}(P)$ for the defender and defender λ^1-LOSS verdicts transfer from P to P'';
(b) every A-complete window of P is A-complete in P'';
(c) a cell empty in P is empty or newly empty in P'', never newly occupied; and
(d) legality may shrink. A deletion argument must therefore satisfy *well-formed legality*: every attacker placement on which it relies lies within distance 8 of a stone not in Y. In the applications below, it lies within distance 3 of an attacker stone, as follows automatically for threat-creating moves from Theorem 3.4, or it lies within distance 8 of a root stone.

Proof. Deleting defender stones can only revive attacker windows, proving (a) and (b). Part (c) follows from Definition 2.4. Part (d) follows from the legality rule in Definition 2.5; the stated witness restores the legality-frontier channel of Lemma 4.1 for every relied-upon cell. □

Completion counting bounds which defender windows can matter before a fixed horizon.

Lemma 4.5 (Completion counting)**.** *If $\mathrm{cnt}_{\mathrm{D}}(W, P) = s$ and the defender makes at most D placements at plies strictly before T, then no play makes W D-complete before ply T unless $s + D \geq 6$.*

Proof. A D-completion needs six defender stones in W. At most s are present in P, and at most D more can be placed before T. □

4.2 Certificate grammar

A certificate records both the searched game tree and verifier-checkable witness data at every way in which that tree can resolve.

Definition 4.6 (Certificate grammar)**.** A *certificate* $\mathcal{C}$ for "A wins from P_0 by horizon T," where T is an absolute ply index, is a finite tree whose nodes carry a position and mover/budget. The verifier checks every clause of the following grammar syntactically.

- The root position is P_0 and is nonterminal. Every node has a path-derived clock equal to the root ply plus its depth. The verifier recomputes this clock. Every resolution label refers to it, and *leaf-ply* is the index of the last placement completed on the path to a leaf.
- Every child is the exact successor, under Definition 2.5, of the placement on its edge. Every placement is legal at its parent. No edge is defender-terminal; a searched defender reply that completes a defender window makes the certificate invalid.
- Every maximal node is typed. An OR node whose designated placement completes is itself an *OR-COMPLETION leaf*. Every internal AND node has a nonempty searched set. At every AND node, and again at every LOSS leaf, the verifier checks that own_win_now is false for the defender.
- An *OR node*, at which A places, has one designated placement. If it completes a window, the node names the witness window W^* and the completion ply. Otherwise it has the successor as its one child.

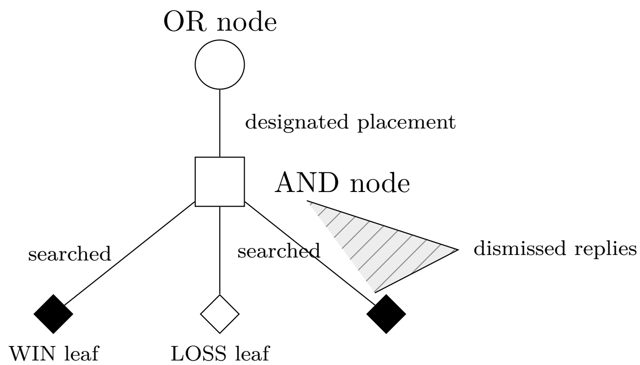


Figure 5: Schematic certificate tree. Solid branches are searched replies; the hatched wedge represents the legal replies covered by a dismissal claim. Circles denote OR nodes, squares denote AND nodes, and diamonds denote typed leaves.

- An *AND node* N, at which D places with budget b, carries a searched set $S(N)$ of legal replies, one child for each searched reply, and a dismissal claim for every other legal reply.
- A *WIN leaf*, at which A is to place, is a verifier-checked own_win_now position. It names an A-alive witness window with count 5 for either budget, or count 4 when $b = 2$. Completion therefore occurs within the attacker's current turn with no intervening defender placement. The leaf names its witness window or windows and a resolution ply at most T.
- A *LOSS leaf*, at which D is to place with budget b and has a λ^1-LOSS, names a witness family $\mathcal{T}$ of A-threat windows whose empty sets have hitting number greater than b. The verifier re-derives this fact and rechecks that own_win_now is false. Its continuation contract is adaptive: for every complete, nonterminal remainder H of the defender's current turn, consisting of exactly b placements unless the game ends earlier, some $W_H \in \mathcal{T}$ survives because its empty-cell set at the leaf is disjoint from H. The attacker then completes the at most two empty cells of W_H during the following turn. The defender places at plies leaf-ply $+ 1, \ldots,$ leaf-ply $+ b$, and the attacker places at leaf-ply $+ b + 1$ and, for a count-4 witness, leaf-ply $+ b + 2$. The declared worst-case resolution ply is leaf-ply $+ b + 2 \leq T$. The leaf names no fixed placement list; the contract quantifies over H.

The horizon T is the maximum resolution ply over the tree. Every λ^1 fact at a leaf is tied to named windows and a checkable continuation contract.

The core collects every cell whose masks or occupancy are used by a future certificate witness.

Definition 4.7 (Certificate core)**.** For a node N, define $\mathrm{core}(\mathcal{C}, N)$ as the union over the subtree rooted at N of

(i) every cell of every named witness window, including OR completions, WIN leaves, and LOSS-leaf families; and

(ii) every attacker placement cell, including leaf-continuation placements.

The protected set adds every defender window that can still complete within the certificate horizon. Its four conditions cover tactical consistency, witness preservation, frontier growth, and attacker legality.

Definition 4.8 (Zone-carrying certificate)**.** At an AND node N with position P_N, put

$$D_N := \mathfrak{D}(P_N, T)$$

and define

$$\mathrm{Prot}(N) = \mathrm{core}(\mathcal{C}, N) \cup \bigcup \left\{ E(W, P_N) : \begin{array}{l} W \text{ is alive for D at } P_N, \\ \mathrm{cnt}_{\mathrm{D}}(W, P_N) + D_N \geq 6 \end{array} \right\},$$

where $E(W, P)$ is the set of empty cells of W in P. For a cell set U, let

$$B(r; U) = \{c : \exists y \in U,\ d(c, y) \leq r\}.$$

The certificate is *zone-carrying* if every AND node N satisfies:

(Z1) $S(N)$ contains the legal empty cells of every A-threat window of P_N. This condition is retained as a redundant safeguard for interior λ^1 consistency: a current threat need not occur in any later witness family, and hitting cells are already the solver's first candidates.

(Z2) $S(N) \supseteq \mathrm{Prot}(N) \cap \mathrm{Legal}(P_N)$. Thus every protected legal cell is searched.

(Z5)

$$S(N) \supseteq \mathrm{Legal}(P_N) \cap B(8D_N; \mathrm{Prot}(N) \setminus (\mathrm{Legal}(P_N) \cup \mathrm{Stones}(P_N)))\,.$$

A defender has at most D_N placements before the horizon, and each can extend legality by at most 8. The horizon-scaled band closes the multi-hop corridor in which successive dismissed stones approach protected territory while later links are ghost-illegal. The band is empty when $\mathrm{Prot}(N) \subseteq \mathrm{Legal}(P_N) \cup \mathrm{Stones}(P_N)$.

(Z4) Every attacker placement in the subtree of N is within distance 8 of an attacker stone of its predecessor position or of a stone of P_0. This condition preserves the legality of attacker placements when defender stones are deleted. It is automatic for threat-creating generators by Theorem 3.4.

When $D_N \leq 5$, both terms of $\mathrm{Prot}(N)$ are finite and verifier-enumerable: a completion-guard window has at least one defender stone and is therefore touched. When $D_N \geq 6$, every all-empty window qualifies. Hence $\mathrm{Prot}(N) \cap \mathrm{Legal}(P_N)$ contains every legal cell in any window alive for D, and the frontier condition also covers its band. The only remaining dismissible cells are deep-interior cells lying in no D-alive window; every window through such a cell is attacker-touched or dead, so the cell can never contribute to a defender completion by Lemmas 4.2 and 4.5. This is the exact boundary of the large-budget case $D_N \geq 6$, not a claim that pruning always vanishes.

Protected cells can disappear under descent, but new protected cells cannot appear.

Proposition 4.9 (Monotonicity of protection)**.** *If M is a descendant AND node of N, then $\mathrm{Prot}(M) \subseteq \mathrm{Prot}(N)$.*

Proof. The core is a subtree union and is monotone under descent. Along the play from N to M, $\mathrm{cnt}_{\mathrm{D}}(W, \cdot)$ increases by the number of intervening defender placements in W, while $\mathfrak{D}(\cdot, T)$ decreases by the number of all intervening defender placements, which is at least as large. Their sum is nonincreasing. A window qualifying at M therefore qualified at N, and its empty set at M is a subset of its empty set at N. □

The frontier guard must also cover dismissed stones that become legal only through earlier dismissed stones.

Lemma 4.10 (Frontier chain closure)**.** In the coupling of Theorem 5.1, let $N(x)$ denote the node at which an X-stone x is introduced. Call x *frontier-clear* if

$$d(x, y) > 8D(N(x)) \quad \text{for every } y \in \mathrm{Prot}(N(x)) \setminus (\mathrm{Legal}(P(N(x))) \cup \mathrm{Stones}(P(N(x)))),$$

where $D(M) = D_M$ and $P(M) = P_M$ for a node M. Then:

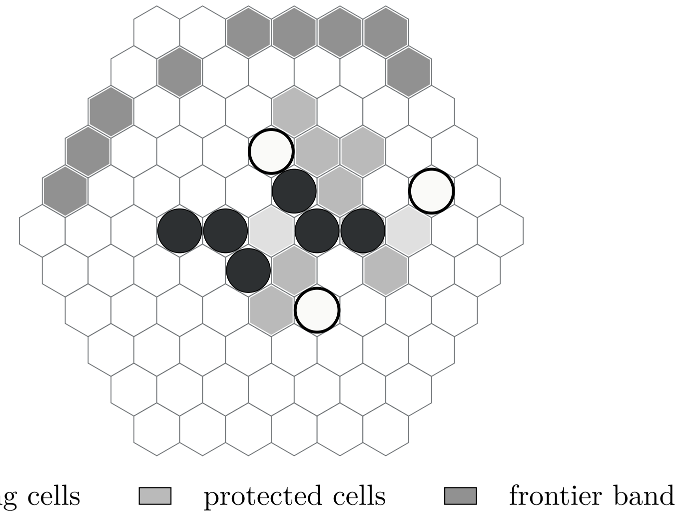


Figure 6: Schematic anatomy of a zone. The light cells hit the central attacker threat; the middle shade represents empties of a named witness window and a completion-guard window; the dark band marks frontier cells. Filled disks are attacker stones and open disks are defender stones.

(a) if every ghost-legal dismissed stone is frontier-clear, as enforced by Item (Z5), then every ghost-illegal dismissed stone is frontier-clear; and
(b) for every AND node N visited by the coupling, no ghost-empty cell of $\mathrm{Prot}(N) \setminus (\mathrm{Legal}(P_N) \cup \mathrm{Stones}(P_N))$ ever becomes legal in the real game while remaining illegal in the ghost.

Proof. Any ghost-empty cell that is real-legal but ghost-illegal has an X-stone as a legality witness. Shared root stones, attacker stones, and searched placements occur in both positions. Cells of Y are ghost-occupied and do not occur in the applications of this lemma. Inductively, the real legality of an X-stone traces through a chain $x_0, x_1, \ldots, x_m$ of X-stones, where x_0 is legal through a shared stone and consecutive links have distance at most 8.

For (a), suppose a ghost-illegal x_j, with $j \geq 1$, is not frontier-clear. There is a protected but not-yet-legal cell y at $N(x_j)$ with $d(x_j, y) \leq 8D(N(x_j))$. The prefix of the chain gives $d(x_0, x_j) \leq 8j$. The stones $x_1, \ldots, x_j$ are later defender placements, so

$$j \leq D(N(x_0)) - 1, \qquad D(N(x_j)) \leq D(N(x_0)) - j.$$

By Proposition 4.9 and monotonic growth of ghost legality, y was also protected but not legal at $N(x_0)$. Therefore

$$d(x_0, y) \leq 8j + 8(D(N(x_0)) - j) = 8D(N(x_0)),$$

contradicting the frontier-clear property of the ghost-legal stone x_0.

For (b), suppose such a protected cell y becomes real-legal while ghost-illegal. It is within distance 8 of the last chain stone x_m. Every defender placement before T has $D(N(x_m)) \geq 1$, so

$$d(x_m, y) \leq 8 \leq 8D(N(x_m)),$$

contradicting part (a). □

The repository proof document states part (b) without the ghost-empty restriction; only the restricted form above is used, in step A3.

The proof will compare the actual play with a certificate walk using canonical difference sets rather than an order relation between positions.

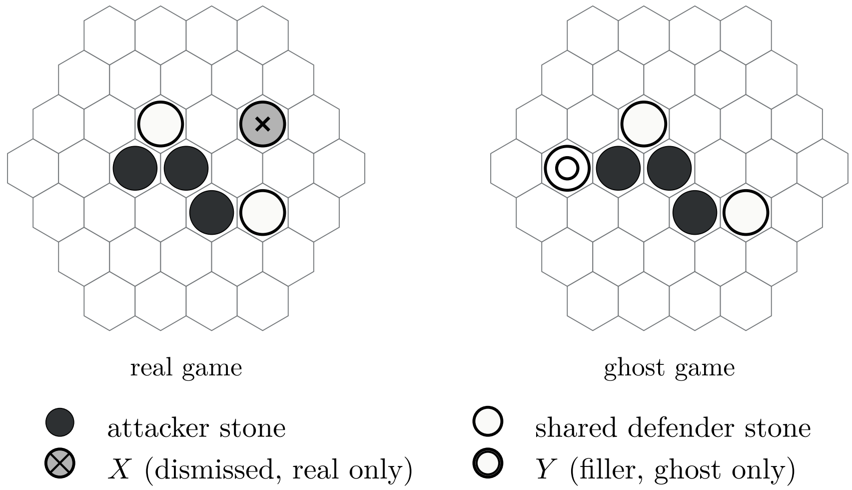


Figure 7: The coupled positions have identical attacker stones and shared defender stones. Canonical difference sets record one real-only dismissed stone and one ghost-only filler.

Definition 4.11 (Real–ghost coupling)**.** The proof of Theorem 5.1 couples a real position R to a ghost position G on $\mathcal{C}$. The positions have the same ply index, mover, and budget, and have identical attacker stones. Define the canonical, disjoint defender-stone differences. For a position $Q = (\sigma_Q, m_Q, b_Q)$, write

$$\begin{aligned}\mathrm{Stones_D}(Q) &:= \{c \in \operatorname{dom}\sigma_Q : \sigma_Q(c) = \mathrm{D}\},\\ X = \mathrm{Stones_D}(R) \setminus \mathrm{Stones_D}(G), &\qquad Y = \mathrm{Stones_D}(G) \setminus \mathrm{Stones_D}(R).\end{aligned}$$

The coupling maintains these invariants at every visited node N:

(i) every $x \in X$ entered at a dismissal step whose node certified $x \notin \mathrm{Prot}$;
(ii) every $y \in Y$ entered as a ghost filler, which is a searched placement of its node;
(iii) $X \cap \mathrm{Prot}(N) = \varnothing$, by part (i) and Proposition 4.9; and
(iv) every $x \in X$ is frontier-clear. Condition Item (Z5) enforces this for ghost-legal dismissals. The chain closure in Lemma 4.10 passes it to ghost-illegal dismissals.

A dismissal-history set $\widehat{X} \supseteq X$ records every cell ever added to X and never removes a cell. It is used when the completion argument must remember an earlier dismissal after the canonical difference has changed.

5 Horizon-parameterized dismissal soundness

The proof runs the attacker's real strategy beside a bookkeeping play, called the ghost, that remains on certificate branches. The two boards have the same attacker stones. The set X contains dismissed defender stones present only in the real game, while Y contains searched filler stones present only in the ghost. The coupling keeps both games at the same ply and budget while tracking the effects of these two differences.

The main result turns the local zone conditions into a strategy against every legal defender reply.

Theorem 5.1 (Horizon-parameterized dismissal soundness)**.** *This statement is proved for valid zone-carrying certificates obeying the grammar of Definition 4.6 and satisfying Items (Z1), (Z2), (Z4) and (Z5), with exact D_N and the full, rather than sharpened, defender-placement budget. Let*

*$\mathcal{C}$ be such a certificate for "*A *wins from P_0 by ply T." Then* A *wins from P_0 against every defender strategy over the full legal move set by a ply at most T. Hence every dismissal in $\mathcal{C}$ is sound.*

Proof. We construct the attacker's real-game strategy by coupling the real position R, initially P_0, to a walk on $\mathcal{C}$ with ghost position G, also initially P_0. We maintain all invariants of Definition 4.11: equal ply, mover, and budget; identical attacker stones; the canonical defender differences X and Y; and the protection and frontier-clear properties. Leaves are processed as soon as the walk reaches them. The induction parameter is T minus the current ply.

Step O (ghost at an OR node, designated placement c). The cell c is empty in the ghost because the certificate is valid. It is not in X: it belongs to the core and therefore to the protected set of every ancestor by Proposition 4.9, while the coupling keeps X out of the protected set. It is not in Y, since cells of Y are ghost defender stones and c is ghost-empty. Thus c is empty in the real position.

The cell is also real-legal. By Item (Z4), it lies within distance 8 of an attacker stone of its predecessor position or a root stone. These stones are identical in R and G. The attacker plays c in both games. The attacker stones remain identical, and X and Y do not change.

If this placement completes the named witness W^* in the ghost, then W^* belongs to the core. Hence $X \cap W^* = \varnothing$. Also $Y \cap W^* = \varnothing$, because an attacker-complete ghost window contains no ghost defender stone. Its masks agree in the two games by Lemma 4.3, so the real placement completes W^* at the same ply, which is at most T.

The real placement may instead complete another window that the ghost does not. Such a window is blocked by Y in the ghost but not in the real game, as characterized by Lemmas 4.3 and 4.4. The real game then terminates immediately with an attacker win before T. If neither completion occurs, the coupling recurses on the child.

Step A (ghost at an AND node N, real defender placement d). The real-legal cell d is either ghost-occupied, ghost-empty and searched, or ghost-empty and unsearched. These cases are exhaustive.

Filler subroutine. This subroutine is used in A2 and A3. The ghost must consume a defender ply. Choose any $d_0 \in S(N)$. This set is nonempty by the certificate grammar, and every member is ghost-legal and ghost-empty. The ghost plays d_0. If d_0 was real-occupied, then $d_0 \in X$ and the update is $X \leftarrow X \setminus \{d_0\}$: the ghost has caught up. Otherwise the update is $Y \leftarrow Y \cup \{d_0\}$. Recomputing the canonical differences preserves the coupling invariants. Every new Y-cell is a searched placement, and X can only shrink.

A1. Suppose d is ghost-empty and $d \in S(N)$. Both games play d, their difference sets do not change, and the ghost descends to the d-child.

A2. Suppose d is ghost-occupied. It is a ghost defender stone: ghost attacker stones also occur in the real game, while d is real-empty. Thus $d \in Y$. The real game plays d, removes it from Y, runs the filler subroutine, and descends to the d_0-child.

A3. Suppose d is ghost-empty and $d \notin S(N)$. This is a dismissal. First, $d \notin \mathrm{Prot}(N)$. If $d \in \mathrm{Legal}(P_N)$, this follows from Item (Z2), since protected legal cells are searched. If $d \notin \mathrm{Legal}(P_N)$ and is real-legal only through the X-expanded frontier, Lemma 4.10 excludes a protected-not-legal cell. The assumption also excludes a protected legal cell.

Second, d is frontier-clear. If $d \in \mathrm{Legal}(P_N)$, a non-clear cell lies in the band of Item (Z5) and would have been searched. If $d \notin \mathrm{Legal}(P_N)$, Lemma 4.10 supplies clearance. The real game plays d and updates

$$X \leftarrow X \cup \{d\}, \qquad \widehat{X} \leftarrow \widehat{X} \cup \{d\}.$$

The protection and clearance invariants hold. The ghost runs the filler subroutine and descends to the d_0-child. This is also where Lemma 4.10 prevents a dismissed-stone corridor from reaching protected frontier territory.

Anchored no-completion argument. We show that the real defender cannot complete a window before ply T. For every window W, the defender stones in the real position are the ghost defender stones, plus X and minus Y. Consequently,

$$\mathrm{cnt}_{\mathrm{D}}(W, R) = \mathrm{cnt}_{\mathrm{D}}(W, G) + |X \cap W| - |Y \cap W| \leq \mathrm{cnt}_{\mathrm{D}}(W, G) + |\widehat{X} \cap W|. \tag{MI}$$

Suppose that W becomes defender-complete in the real play at a ply less than T.

If no placement in W was ever dismissed, then $\widehat{X} \cap W = \varnothing$. Inequality (MI) gives $\mathrm{cnt}_{\mathrm{D}}(W, G) \geq 6$ at that ply. The ghost follows an actual certificate line, but the grammar of Definition 4.6 forbids defender-terminal edges and requires exact successors. No valid line can contain this completion, a contradiction.

Otherwise let x^* be the first cell of W ever placed into $\widehat{X}$, and let N^* be its dismissal node. Write $P(N^*)$, $R(N^*)$, and $D(N^*)$ for the ghost position, real position, and remaining defender count at that node. Step A3 gives $x^* \notin \mathrm{Prot}(N^*)$. The window W is alive for the defender at N^*: it completes later, and defender-aliveness persists backward by Lemma 4.2. The cell x^* is one of its empties. If

$$\mathrm{cnt}_{\mathrm{D}}(W, P(N^*)) + D(N^*) \geq 6,$$

then W qualifies for the completion guard and all its empties, including x^*, lie in $\mathrm{Prot}(N^*)$, a contradiction. Hence the displayed sum is less than 6.

Before x^*, no cell of W belongs to $\widehat{X}$, so (MI) gives

$$\mathrm{cnt}_{\mathrm{D}}(W, R(N^*)) \leq \mathrm{cnt}_{\mathrm{D}}(W, P(N^*)).$$

The plays are synchronized. From x^* onward, at most $D(N^*)$ real defender placements occur strictly before T. Applying Lemma 4.5 at N^* shows that W cannot reach count 6 before T, again a contradiction.

Leaf transfer. Post-leaf plies are analyzed directly in the real game; the coupling is not extended past the leaf.

At a WIN leaf, Definition 4.6 provides a checked own_win_now witness with count 5, or with count 4 and both remaining attacker placements. No defender placement is interleaved. The witness window belongs to the core, so it is disjoint from X; it is defender-free in the ghost and therefore disjoint from Y. Its masks agree by Lemma 4.3. Its empty cells lie within distance 2 of shared attacker stones by Lemma 3.1, so they are legal in the real game. They are also real-empty: X avoids the core and Y-cells are absent from the real position. The attacker fills the cells and completes the window at the declared resolution ply, at most T.

At a LOSS leaf with defender budget b, the named family $\mathcal{T}$ has hitting number greater than b. The real and ghost masks and empty sets of every $W \in \mathcal{T}$ are identical at the leaf. Such windows belong to the core, so X avoids them. They are attacker-alive in the ghost, so they contain no ghost defender stone and in particular no Y-cell. Attacker stones are shared. Thus the real hitting number of $\mathcal{T}$ is the same as the ghost hitting number and is greater than b.

It remains to exclude a real defender own_win_now at the leaf. Let W be a putative witness. If $\widehat{X} \cap W = \varnothing$, the leaf-time form of (MI) gives $\mathrm{cnt}_{\mathrm{D}}(W, R) \leq \mathrm{cnt}_{\mathrm{D}}(W, G)$. The attacker mask and budget agree, so the real predicate would imply the ghost predicate. This contradicts the verifier check required at the leaf. If $\widehat{X} \cap W \neq \varnothing$, the second case of the anchored argument, applied at the first dismissed cell $x^* \in W$, is pure counting over every real defender placement before T. It does not require the coupling to continue. It forbids W from reaching count 6 before T. Yet a current-turn win would complete before the declared resolution, which is at most T. This is also a contradiction.

Now let the real defender play any complete remainder H of the current turn. Since $|H| \le b$ is smaller than the hitting number of $\mathcal{T}$, some $W_H \in \mathcal{T}$ satisfies $E(W_H) \cap H = \varnothing$. The window remains attacker-alive, because its leaf masks agreed with the ghost and its cells avoid H. It has at most two empty cells. By Lemma 3.1, they lie within distance 2 of shared, permanent attacker stones and remain legal and empty in the real game. The attacker completes W_H during the following turn, using one placement for a count-5 witness and two for a count-4 witness. Completion occurs by

$$\text{leaf-ply} + b + 2 \le T.$$

The set H may contain hits on other witness windows, quiet moves, or frontier moves. The argument uses only its cardinality and permanence. This completes the induction. □

6 Computable relevance zones

The soundness theorem can be read as an explicit searched-set formula at each AND node. Its terms separate current tactics, certificate witnesses, defender completions, and frontier reachability.

Theorem 6.1 (Zone reading)**.** *This statement has the same boundary as Theorem 5.1: it applies to valid zone-carrying certificates obeying Definition 4.6 and satisfying Items (Z1), (Z2), (Z4) and (Z5), with exact D_N and the full, rather than sharpened, defender-placement budget. At an AND node, the sound searched set is*

$$Z(N) = \mathrm{Legal}(P_N) \cap \big[\,\mathrm{hitting}(P_N) \cup \mathrm{Prot}(N) \\ \cup B\big(8D_N; \mathrm{Prot}(N) \setminus (\mathrm{Legal}(P_N) \cup \mathrm{Stones}(P_N))\big)\big],$$

where $\mathrm{hitting}(P)$ *is the union of the empty sets of the* A*-threat windows of P, namely the cells in the members of* $\mathrm{hit}(P)$*. The three terms are the hitting cells, the protected cells consisting of the core and completion-guard empties, and the horizon-scaled frontier band.*

The completion guard counts all defender plies before the horizon. The sharpened version that counts only non-hit placements is not proved; it is listed in Section 11. The frontier band is empty whenever $\mathrm{Prot}(N) \subseteq \mathrm{Legal}(P_N) \cup \mathrm{Stones}(P_N)$*. This is the typical case, since witness windows cluster near the action.*

Proof. The three terms are exactly the searched cells required by Items (Z1), (Z2) and (Z5). With Item (Z4) and the certificate grammar, Theorem 5.1 proves that dismissing every other legal reply is sound. □

For short threat-creating continuations, future attacker placements remain inside windows already touched by current attacker stones.

Lemma 6.2 (Short-range attacker-core containment)**.** *Suppose every attacker placement in a certificate is threat-creating: it lands in a window that, immediately before the placement, is alive for* A *and contains at least three attacker stones. If a placement is the attacker's kth future placement from a node N, then for $k \le 3$ its window contains at least*

$$3 - (k-1) \ge 1$$

attacker stones already present in P_N. Thus the placement cell lies in an A*-touched window that is alive for* A *at P_N. This statement covers attacker-placement core cells; named witness windows are a separate, certificate-known component of the core.*

Proof. The window has at least three attacker stones at placement time. At most $k-1$ of them can be future stones, one from each earlier future attacker placement. At least $3-(k-1)$ are therefore current at N, which is positive for $k \le 3$. Since the window is alive immediately before the placement, backward permanence from Lemma 4.2 makes it alive at P_N. □

From the fourth future attacker placement onward, an attacker core cell may lie in a currently virgin window. Such setup cells must be supplied explicitly from the certificate core.

A common radius-three generator covers the full zone only under a joint set of short-horizon and witness-location hypotheses.

Theorem 6.3 (Static-set coverage for short horizons)**.** This is a coverage theorem under all of the following joint hypotheses: $D_N \le 3$; at most three remaining attacker placements are made; those placements are threat-creating; every named witness-window cell is in an A-touched window that is alive for A at P_N; and the frontier band is empty. Let

$$r_3(P) = \{c \in \mathrm{Legal}(P) : \exists s \in \mathrm{Stones}(P),\ d(c,s) \le 3\}.$$

Then

$$r_3(P_N)\ \cup\ \{\text{empty cells of A-touched alive windows at } P_N\}$$

covers $Z(N)$.

Without all of these hypotheses, the searched set must include $\mathrm{core}(\mathcal{C}, N) \cap \mathrm{Legal}(P_N)$ explicitly. The static set is then a candidate generator, not a certified zone by itself.

Proof. For $D_N \le 3$, a completion-guard window has at least three defender stones, so each of its empty cells is within distance 3 of a defender stone by Lemma 3.1. Hitting cells are within radius 2 by Theorem 3.3. Under the stated limit on the remaining threat-creating placements, Lemma 6.2 puts attacker-placement core cells in A-touched alive windows. The separate witness hypothesis does the same for named witness cells, and the last hypothesis removes the frontier term. These observations cover every term of the formula in Theorem 6.1. □

When every defender turn is already fully occupied by threat hits, the completion and frontier guards can be removed through a local auxiliary refutation.

Theorem 6.4 (Forced-tree regime)**.** Let $\mathcal{C}$ be a valid certificate under Definition 4.6. Suppose every internal AND node with defender budget b has verifier-checked $\mathrm{mhs}(P_N) = b$, and the defender's immediate-win predicate is false. Then the sufficient searched set is

$$S(N) = \mathrm{hitting}(P_N) \cup \big(\mathrm{core}(\mathcal{C}, N) \cap \mathrm{Legal}(P_N)\big).$$

The completion-guard and frontier terms are unnecessary. The auxiliary refutation used in the proof is measured against the same certified horizon T.

Proof. Work at the first dismissed reply d. Before it, the real play follows certificate lines exactly. Let b be the defender budget at its node. Since every hitting cell is searched, d is non-hitting. Attach an auxiliary refutation according to b.

(a) If $b = 2$, the successor remains a defender node with budget $b' = 1$. The untouched threat family still needs two hits, so its hitting number is greater than b'. It supplies a defender LOSS leaf with the current threat family as its witness family.

(b) If $b = 1$, the successor has the attacker to move with budget 2 and at least one surviving attacker threat. This supplies an attacker WIN leaf: a count-4 witness at budget 2, or a count-5 witness, satisfies own_win_now.

The auxiliary leaf witnesses are the node's current attacker-threat windows. Their empty cells are hitting cells of the node and are covered by the searched set. Thus the auxiliary branch needs no additional appeal to the original subtree's core.

Its resolution fits inside the original horizon T. Had the defender used a searched minimum hitting reply instead, a pair when $b = 2$ or a common hit when $b = 1$, every current attacker threat would be killed; defender placements cannot create a new attacker threat. The original certificate therefore cannot resolve on the first following attacker ply. Its horizon already reaches the second attacker ply, which is exactly what the auxiliary branch needs: leaf-ply + $b' + 2$ in the $b = 2$ case, and the attacker's own turn in the $b = 1$ case.

It remains to rule out defender completion before the auxiliary resolution. Because the defender's immediate-win predicate is checked as false at each AND node, a defender-alive window has defender count at most 3 when $b = 2$, and at most 4 when $b = 1$. Indeed, count 4 at budget 2, or count 5 at either budget, would satisfy own_win_now. When $b = 2$, the dismissal and the single placement in the $b' = 1$ LOSS contract contribute two defender placements in total; the latter placement is counted once. The count reaches at most $3 + 2 = 5$. When $b = 1$, only the dismissal precedes the attacker's turn, so the count reaches at most $4 + 1 = 5$. No defender window completes before the attacker resolution. $\square$

6.1 The relevance closure

The implementable closure uses only touched-window queries and falls back to the full legal set when a finite touched-window representation no longer covers every completion guard.

Definition 6.5 (Horizon relevance closure)**.** Let P have the defender to move, and let $D = \mathfrak{D}(P, T)$ be the number of defender placements before the horizon. Define

$$\mathcal{A}(P) = \{\text{empties of alive windows with } \mathrm{cnt_A} \geq 1 \text{ and } \mathrm{cnt_D} = 0\}$$

and

$$\mathcal{G}(P, D) = \{\text{empties of D-alive windows with } \mathrm{cnt_D} \geq 6 - D\}.$$

Then

$$R(P, D) = \begin{cases} \mathrm{Legal}(P) \cap [\mathrm{hitting}(P) \cup \mathcal{A}(P) \cup \mathcal{G}(P, D)], & D \leq 5, \\ \mathrm{Legal}(P), & D \geq 6. \end{cases}$$

For $D \leq 5$, $\mathcal{G}(P, D)$ is a finite touched-window query.

The set R is a candidate generator, not a certified zone by itself. The set $\mathcal{A}$ covers attacker core cells only within the range of Lemma 6.2; deeper setup cells may lie outside every term of R. Certified sufficiency uses

$$R(P_N, D_N) \cup (\mathrm{core}(\mathcal{C}, N) \cap \mathrm{Legal}(P_N)) \cup \big[\mathrm{Legal}(P_N) \cap B\big(8D_N; \mathrm{Prot}(N) \setminus (\mathrm{Legal}(P_N) \cup \mathrm{Stones}(P_N))\big)\big].$$

The extra terms are certificate-known and verifier-enumerable. The $D \geq 6$ fallback is conservative. The soundness theorem can still dismiss deep-interior cells at that depth when they lie in no defender-alive window and are outside the hitting, core, and frontier terms. Lying in no defender-alive window is necessary but not sufficient for dismissal. Every certified-sufficiency claim also presupposes a valid certificate under Definition 4.6 and a separate verification of Item (Z4).

The closure plus the two certificate-derived additions covers each verifier-required zone term.

Theorem 6.6 (Closure and certificate-extras coverage)**.** This is a coverage theorem. Searching at every AND node the set

$$R(P_N, D_N) \cup (\mathrm{core}(\mathcal{C}, N) \cap \mathrm{Legal}(P_N)) \\ \cup \big[\mathrm{Legal}(P_N) \cap B\big(8D_N; \mathrm{Prot}(N) \setminus (\mathrm{Legal}(P_N) \cup \mathrm{Stones}(P_N))\big)\big]$$

satisfies Items (Z1), (Z2) and (Z5). Certified dismissal soundness additionally presupposes a valid certificate under Definition 4.6, separate verification of Item (Z4), exact D_N, and the full defender-placement budget.

Both certificate terms are verifier-enumerable. The legal part of the core is usually redundant, for example when the next designated attacker move is covered by Lemma 6.2, rather than empty. The band is empty whenever all protected territory is already legal or occupied. Computing R takes one window-store pass per node, or $O(\#\text{touched windows})$ time.

Proof. The hitting term of R gives Item (Z1). For $D_N \leq 5$, its guard term contains the completion-guard part of $\mathrm{Prot}(N)$, while $\mathcal{A}(P_N)$ covers the short-range attacker cells described by Lemma 6.2; the explicit legal core supplies every remaining protected certificate cell. For $D_N \geq 6$, R is the full legal set. The displayed band is exactly the set required by Item (Z5). Hence the union satisfies the three stated conditions, and Theorems 5.1 and 6.1 give sound dismissal once the remaining hypotheses are verified. ☐

The practical solver loop uses R as a heuristic candidate generator. When a proof is found, the verifier recomputes Items (Z1), (Z2), (Z4) and (Z5) exactly at every node against the actual certificate, including its core and frontier band. A violation yields UNKNOWN. Search trimming can therefore affect completeness, but it cannot create a false verified win.

On random midgame positions, the generator has measured a mean of 147 cells against 302 legal cells, with smaller sets in forced regions. In the fully-forcing regime of Theorem 6.4, the defender set consists of hitting cells and the certificate's own legal core cells at any depth. For $D \geq 6$, the completion guard covers every legal cell in a defender-alive window, leaving only interior cells in no such window as possible dismissals, subject also to the hitting, core, and frontier conditions. This $D \geq 6$ boundary is a theorem rather than an experimental observation.

7 Impossibility of global pairing

A pairing strategy preassigns disjoint pairs and answers each attacker stone in a paired cell by taking its mate. In a single-placement game this gives a perfect local defense. Under Hexo's two-placement turns, the attacker may occupy both cells of a pair within one turn, so even an existing pairing would need a separate argument. The theorem below makes the question moot: the required matching does not exist.

Theorem 7.1 (No pairing; source T8)**.** *There is no partial matching M on the cells, with each cell in at most one pair, such that every window contains both cells of some pair.*

Proof. Discard every pair that is not contained in a window. This removes every non-axis-aligned pair and every axis pair at distance greater than 5, while preserving the covering property. By fact 2.3, each remaining pair is axis-aligned at an axis-distance $\delta \in \{1, \ldots, 5\}$.

Fix a lattice line ℓ. A pair at positions $(p, p+\delta)$ is contained in the windows whose integer starts satisfy

$$s \in [p + \delta - 5, p].$$

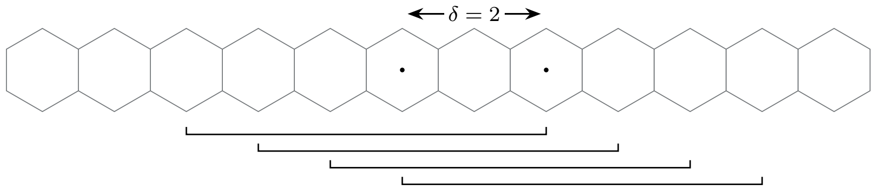


Figure 8: A pair at axis-distance $\delta = 2$ is contained in exactly $6 - \delta = 4$ length-six windows. The four brackets show their possible starts.

There are exactly $6-\delta \le 5$ such starts. If the left cells of consecutive pairs are p_i and p_{i+1}, coverage of every start requires $p_{i+1} - p_i \le 6 - \delta_{i+1} \le 5$. Allowing mixed values of δ cannot improve this bound. A segment of ℓ with L cells has $L-5$ internal window starts and therefore contains at least $(L-5)/5$ pairs. At least $2(L-5)/5$ of its cells are matched along the axis of ℓ.

Each matched cell serves exactly one axis, namely the axis of its pair. Let

$$B_R = \{x : d(x,0) \le R\}, \qquad |B_R| = 3R^2 + 3R + 1.$$

For each of the three axes, B_R meets exactly $2R+1$ line segments, and the lengths of those segments sum to $|B_R|$. Summing the line bound over all segments and all three axes gives

$$\text{matched-cell incidences} \;\ge\; \frac{6}{5}|B_R| - 6(2R+1).$$

A matching supplies at most $|B_R|$ incidences. At $R = 20$, however,

$$\frac{6}{5}(1261) - 246 = 1267.2 > 1261,$$

which is a contradiction. $\square$

The same density calculation applies to other translation-invariant line systems with negligible ball boundary.

Corollary 7.2 (Translation-invariant line grids; source T8.1)**.** *Consider k-in-a-row on a point lattice with g pairwise nonparallel, translation-invariant line directions. Suppose every lattice line has a length-k window at every offset and the lattice balls have boundary-to-volume ratio tending to zero, as in the Følner property of $\mathbb{Z}^2$. A pairing requires per-line matched density $2/(k-1)$ and hence*

$$g\frac{2}{k-1} \le 1 \qquad \Longrightarrow \qquad k \ge 2g+1.$$

For the square grid, $g = 4$ gives $k \ge 9$, consistent with the classical existence of pairing draws for 9-in-a-row. For the hex grid, $g = 3$ gives $k \ge 7$, so 6-in-a-row misses the bound by one. Existence at the threshold requires a separate construction and is not claimed. The conclusion applies to lattices satisfying the stated hypotheses, not to arbitrary grids.

The global-pairing approach is therefore closed. Partial pairings on constrained regions are not ruled out, but they cannot be position-independent.

8 Potential-function certificates

The potential method follows Erdős and Selfridge's hypergraph criterion and its biased-game form developed by Beck [3, 4]. Here it supplies sufficient blocking certificates for selected Hexo windows. Window births prevent the direct global invariant from giving an unrestricted forever theorem.

8.1 Potential and the round inequality

The potential counts only attacker-touched windows that remain available to the attacker.

Definition 8.1 (Residual potential; companion D3 and D4)**.** Let F be a finite labelled family of attacker target windows. At a position P, write A and D for the sets of cells occupied by attacker and defender stones, respectively, and put

$$\mathcal{L}_F(P) = \{W \in F : W \cap D = \varnothing\}, \qquad E_P(W) = W \setminus A,$$

and set

$$\lambda = \sqrt{3}, \qquad \Psi_F(P) = \sum_{W \in \mathcal{L}_F(P)} \lambda^{-|E_P(W)|}.$$

The dynamic potential is

$$\Phi(P) = \sum_{\substack{W:\, W \cap A \neq \varnothing \\ W \cap D = \varnothing}} \lambda^{-e_P(W)}, \qquad e_P(W) = |W \setminus A|.$$

Thus Φ includes exactly the attacker-touched alive windows. All-empty windows are excluded: including them would add $\lambda^{-6} = 1/27$ for infinitely many windows and make the sum diverge.

For an empty cell x, define its fixed-family danger by

$$d_F(x; P) = \sum_{\substack{W \in \mathcal{L}_F(P) \\ x \in E_P(W)}} \lambda^{-|E_P(W)|}.$$

Sequential greedy defense recomputes this danger before each of the defender's two placements and chooses a maximum-danger empty cell, using fixed deterministic tie-breaking. When every scored danger is zero, the filler chooses an occupied cell of maximum q, breaking ties by maximum r, and places at its neighbor one step in the positive Q direction.

The value $\lambda = \sqrt{3}$ is the largest, and hence optimal, base supported by the two-defender, two-attacker round calculation.

Lemma 8.2 (One full fixed-family round; companion L5)**.** *Let the defender make two sequential greedy placements in a fixed residual family, followed by at most two attacker placements. Assume either that all positive-danger cells are legal or that every surviving label is attacker-alive. The fixed-family potential at the next defender epoch is no larger than at the preceding defender epoch.*

Proof. Let δ_1 and δ_2 be the exact reductions made by the two greedy defender placements. After both placements, let

$$S = \max_x d_F(x)$$

over the remaining empty cells. Deleting edges never increases a cell's danger. The first and second greedy maxima are therefore both at least S, so

$$\delta_1 + \delta_2 \geq 2S. \tag{1}$$

An attacker placement at x multiplies by λ the weight of each residual edge containing x, and hence increases the potential by exactly $(\lambda - 1) d_F(x)$. After one attacker placement every remaining danger is at most λS. The two attacker placements therefore increase the potential by at most

$$(\lambda - 1)(1 + \lambda) S = (\lambda^2 - 1) S = 2S. \tag{2}$$

Equations (1)–(2) show that the full round change is nonpositive. If play stops after the first attacker placement, omitting the second nonnegative increase only strengthens the inequality up to that placement. □

Base 2 does not support the claimed blocking threshold. Take three target windows so far apart that no length-six window intersects two of them, and put four attacker stones in each target. Each target then has weight 2^{-2}, so the sum is $3/4 < 1$. The defender can kill at most two separated targets in one turn. The attacker fills the two empties of the survivor and wins.

8.2 Certificates and their boundary

The fixed-family theorem applies only at the start of a full defender turn.

Theorem 8.3 (Fixed-family Hexo blocking; companion Theorem 1)**.** *Let P_0 be a nonterminal defender-*FirstStone *position, and let F be any finite fixed family of windows that are attacker-alive at P_0. If*

$$\Psi_F(P_0) < 1,$$

then sequential greedy defense prevents the attacker from ever completing a window in F.

Proof in the companion. The round inequality keeps $\Psi_F < 1$ at every defender epoch and through the relevant prefixes of the following attacker pair, whereas a completed target has a term of weight 1. The full-turn premise is used for both greedy reductions. With only one defender placement remaining, two separated count-4 targets have potential $2/3 < 1$, but one survives and is completed. The condition is sufficient only; one defender stone may kill as many as 18 windows, so necessity is false. The complete proof is in [5]. □

For the dynamic family, newly touched windows must be enrolled after they are born.

Definition 8.4 (Defender epochs and birth mass; companion D5)**.** Let P_t be the position at the start of the defender's tth full turn. During the following attacker pair, or its prefix before a terminal placement, let N_t consist of the windows that were stone-free at P_t, were not hit by the intervening defender pair, and first became attacker-alive during that attacker pair. For $W \in N_t$, let $a_t(W) \in \{1, 2\}$ be the number of stones from that pair in W at the end of the pair or terminal prefix. Its exact birth mass is

$$C_t = \sum_{W \in N_t} \lambda^{-(6-a_t(W))}.$$

After a terminal play, all later source terms are zero.

The potential below one does certify the next complete attack cycle.

Theorem 8.5 (Global one-cycle certificate; companion Theorem 2)**.** *If the defender is at* FirstStone *and the current dynamic potential satisfies $\Phi(P_t) < 1$, then two sequential touched-window greedy placements prevent the attacker from winning during the immediately following two-placement turn.*

Proof in the companion. Theorem 8.3 blocks every currently alive window, including a completion on the first attacker placement. A different potential winning window either already has a permanent defender stone or was stone-free at P_t and cannot receive six stones in one attacker pair. This is not a forever certificate, because births may put $\Phi(P_{t+1})$ above 1. The complete proof is in [5]. □

A forever statement requires a bound that is uniform over attacker strategies, not a source sum observed along one path.

Theorem 8.6 (Cumulative-birth forever certificate; companion Theorem 3)**.** *Fix the tie-breaking and filler policy in Definition 8.1, and use sequential touched-window greedy defense from P_0. Suppose a uniform constant B_∞ satisfies*

$$\sup_\tau \sum_{t=0}^{\infty} C_t(\tau) \le B_\infty, \qquad \Phi(P_0) + B_\infty < 1,$$

where the supremum is over every legal attacker strategy τ. Then the defender prevents the attacker from ever completing a window. More generally, if B_T is a uniform upper bound satisfying

$$\sup_\tau \sum_{t=0}^{T-1} C_t(\tau) \le B_T, \qquad \Phi(P_0) + B_T < 1,$$

then the first T attacker pairs, indexed $0, \ldots, T-1$, are certified.

Proof in the companion. The dynamic recurrence $\Phi(P_{t+1}) \le \Phi(P_t) + C_t$ and induction keep the potential below one before each certified pair, so Theorem 8.5 applies repeatedly. The supremum is used to make the same argument against every attacker strategy; a pathwise observed value of $\sum_t C_t$ has the wrong quantifier and is not a certificate. The complete proof is in [5]. □

Finite confinement supplies a computable reservoir of virgin windows.

Definition 8.7 (Finite-region targets; companion D6)**.** Fix a defender epoch P_0 and a finite set R of cells empty at P_0. Assume every future attacker placement is confined to R. Define

$$\mathcal{A}_R = \{W : W \cap D_0 = \varnothing,\ W \cap A_0 \neq \varnothing,\ W \setminus A_0 \subseteq R\},$$
$$\mathcal{V}_R = \{W : W \text{ is stone-free at } P_0,\ W \subseteq R\}, \qquad N_R = |\mathcal{V}_R|,$$

and

$$\Phi_R(P_0) = \sum_{W \in \mathcal{A}_R} \lambda^{-e_{P_0}(W)}.$$

Region-target greedy scores only currently attacker-alive members of the fixed target family $\mathcal{H}_R = \mathcal{A}_R \cup \mathcal{V}_R$.

Definition 8.8 (Virgin activation reserve; companion D7 and Lemma 9)**.** For a matching M of pairwise vertex-disjoint two-cell subsets of R, let

$$q(M) = |\{W \in \mathcal{V}_R : \text{ some } \{x, y\} \in M \text{ satisfies } x, y \in W\}|,$$

and put

$$P_R = \max_M q(M), \qquad B_R^* = N_R \lambda^{-5} + P_R(\lambda^{-4} - \lambda^{-5}).$$

For certification, P_R must be evaluated exactly or replaced by a proved upper bound, such as $P_R \le \min\{N_R, 5\lfloor |R|/2 \rfloor\}$. An exhibited matching gives only a lower bound and is unsafe here. The coarser reserve is $B_R^* \le N_R/9$.

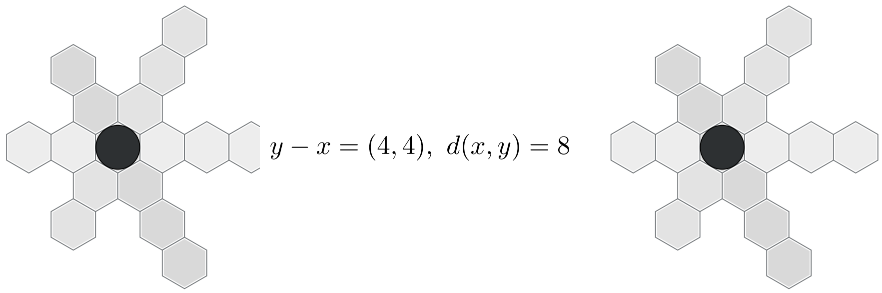


Figure 9: Counterexample 1, schematically. The two attacker placements are at hex distance 8, off all three axes, and have disjoint clean window stars. Together they enroll 36 one-stone windows, and the potential jumps from 0 to $36\lambda^{-5} = 4/\sqrt{3} > 1$. Three of the 18 windows through each new stone are shaded.

Theorem 8.9 (Finite-region forever theorem; companion Theorem 4)**.** *If the attacker is forever confined to the finite region R and*

$$\Phi_R(P_0) + B_R^* < 1,$$

then region-target greedy defense prevents the attacker from ever winning. The weaker sufficient condition

$$\Phi_R(P_0) + \frac{N_R}{9} < 1$$

has the same conclusion. Every positive-reduction defender placement made by this strategy is an empty cell of a currently attacker-alive member of $\mathcal{H}_R$ and is legal. The confinement is on the attacker; defender fillers may lie outside R.

Proof in the companion. Delayed enrollment gives the fixed-target recurrence, while the matching reserve charges each virgin window once and charges the larger two-stone birth weight to at most P_R windows. Every window completable under confinement belongs to $\mathcal{A}_R$ or $\mathcal{V}_R$. The confinement premise must come from a separate certificate; the theorem does not establish it. The complete proof is in [5]. □

8.3 Practical domain and open boundary

The exact node test can be performed without floating-point arithmetic.

Corollary 8.10 (Integer-only threshold check; companion Corollary 2)**.** *At a nonterminal position, for $s = 1, \ldots, 5$, let n_s be the number of currently attacker-alive windows with exactly s attacker stones. Put*

$$a = n_1 + 3n_3 + 9n_5, \qquad b = n_2 + 3n_4.$$

Then

$$\Phi < 1 \iff b \le 8, \qquad a^2 < 3(9-b)^2.$$

Indeed, $\Phi < 1$ is $a + \sqrt{3}b < 9\sqrt{3}$. The right side of $a < \sqrt{3}(9-b)$ must be positive, giving $b \le 8$, after which squaring is equivalent. This check costs $O(\#\text{touched windows})$, not $O(\#\text{threats})$.

If $I = \sum_{s=1}^{5} sn_s$ is the number of attacker-stone/alive-window incidences, then $\Phi \ge I/18$. Thus $\Phi < 1$ forces $I \le 17$. With N_A attacker stones, all but at most 17 of the $18N_A$ incidences must already lie in defender-killed windows. The practical domain is therefore a heavily blocked, endgame-like region rather than a general middlegame.

The raw global claim that $\Phi < 1$ blocks the unrestricted infinite game forever remains open. Counterexample 1 starts with $\Phi = 0$ and, after the fixed zero-danger fillers, uses two far placements to birth 36 one-stone windows, producing $4/\sqrt{3} > 1$ as shown in Fig. 9. Requiring that no current attacker window have at most two empties does not repair the invariant, because the condition holds vacuously at $\Phi = 0$ before the same births. Replacing $\Phi < 1$ by $\Phi \leq 1$ also fails: three separated count-4 targets have total potential 1, the defender kills at most two, and the attacker completes the survivor. These are failures of the proposed potential certificates, not proofs that the attacker wins every corresponding unrestricted game. The $\Phi = 1$ example is a static blanket-game construction; it is not asserted to be a material-balanced reachable engine position and does not prove sharpness after restriction to reachable nodes.

Every positive-reduction greedy placement lies in an attacker-touched alive window and therefore inside the relevance closure of Section 6. If no monitored window remains, one or both placements of a defender turn may instead be arbitrary legal fillers. The certified strategy must permit these fillers. Their presence does not affect the dismissal soundness theorem; it only constrains the potential-certified strategy.

9 Domination patterns

Domination compares a searched defender reply with an omitted reply over a finite stopped horizon. The comparison transfers complete strategies and checks occupancy, window masks, and the legality frontier after every mapped placement. This prevents local mask similarity from hiding a different legal continuation.

For a cell x, write $B_R(x) = \{z : d(x, z) \leq R\}$ and let $\Omega(x)$ be the 18 windows containing x. For a post-opening position P, define its geometric support by

$$\Lambda(P) = \bigcup_{s \in \mathrm{Stones}(P)} B_8(s).$$

If P is nonterminal, its legal placements are $L(P) = \Lambda(P) \setminus \mathrm{Stones}(P)$; if P is terminal, $L(P) = \varnothing$. An empty cell is *dead* when every window in $\Omega(x)$ is dead.

9.1 Stopped outcomes and causal simulation

The outcome records the first terminal placement, with the defender's reply itself at time zero.

Definition 9.1 (Stopped horizon outcome; companion D3 and D4)**.** Let S be a possibly terminal successor of a defender reply. A horizon n allows at most n further placements after that reply. Given pure continuation strategies σ_D and τ_A, play stops at the first terminal placement or after those n placements, and

$$\mathrm{Out}_n(S, \sigma_D, \tau_A) \in \{D@t : 0 \leq t \leq n\} \cup \{A@t : 0 \leq t \leq n\} \cup \{?\}.$$

Here ? means that no terminal placement occurred through the horizon. For defender comparison, set

$$u_D(D@t) = 1, \qquad u_D(?) = 0, \qquad u_D(A@t) = -1.$$

Winning windows and times remain in the trace, although all wins by one player have the same qualitative value. A pure X-strategy assigns a legal placement to every nonterminal history of length less than n at which X is to place, including both consecutive histories of an X turn, and is never queried after a terminal history. Write $\Sigma_X(S)$ for the set of such strategies from S.

The transfer directions encode the defender-pruning implication.

Definition 9.2 (n-outcome domination; companion D5 and Lemma 3)**.** Let a and b be legal defender replies at the same nonterminal position P, with stopped successors $S_a = P + a$ and $S_b = P + b$. The reply b is *n-outcome-dominated by a for the defender*, written $b \preceq_n a$, if there are maps

$$T_D : \Sigma_D(S_b) \to \Sigma_D(S_a), \qquad T_A : \Sigma_D(S_b) \times \Sigma_A(S_a) \to \Sigma_A(S_b)$$

such that, for every discarded-branch defender strategy σ_b and every searched-branch attacker strategy τ_a,

$$u_D(\mathrm{Out}_n(S_a, T_D(\sigma_b), \tau_a)) \geq u_D(\mathrm{Out}_n(S_b, \sigma_b, T_A(\sigma_b, \tau_a)))\,.$$

Equivalently,

$$\forall\sigma_b\ \exists\sigma_a\ \forall\tau_a\ \exists\tau_b : \quad u_D(\mathrm{Out}_n(S_a, \sigma_a, \tau_a)) \geq u_D(\mathrm{Out}_n(S_b, \sigma_b, \tau_b)).$$

Consequently, an attacker win forced by the horizon after the searched reply a is also forced after the omitted reply b.

A causal simulation is the checkable witness used by the patterns below.

Definition 9.3 (Causal alternating simulation; companion D6)**.** A depth-k defender-favouring alternating simulation relates a searched state G to an omitted state H at equal placement depth. A defender-terminal G, or an attacker-terminal H, is accepted immediately. An attacker-terminal G is accepted only if H is already attacker-terminal; a defender-terminal H is accepted only if G is already defender-terminal. No move is generated after a terminal state. Two nonterminal states are accepted when $k = 0$.

Otherwise the states have the same player to move and the same remaining placement budget. At a defender history, every legal move in H is mapped causally to a legal move in G. At an attacker history, every legal move in G is mapped causally to a legal move in H. The successor pair is related at depth $k - 1$. The map depends only on the paired prefix and current move. It pairs FirstStone with FirstStone and SecondStone with SecondStone, including stored first-cell coordinates under any coordinate involution.

Such a simulation implies n-outcome domination by recursively transferring the two players' strategies [5]. Each simulation step must account for three channels: the mapped cells must be empty with explicit images for branch-exclusive empties; after each mapped placement, every labelled mask in the union of the two incident 18-window sets must be compared, with the terminal clauses applied before phase advance; and legal moves must be included in the actor-dependent direction.

9.2 Dead cells and interchangeable hits

A dead empty has no legality frontier because the dead-window witnesses on the six directed rays already support its whole radius-eight ball.

Lemma 9.4 (A dead empty has no legality frontier; companion Lemma 7)**.** *Let b be empty and dead in a nonterminal post-opening position P. Then*

$$B_8(b) \subseteq \Lambda(P).$$

In particular, b is legal, and placing at b inserts no previously unsupported cell into the legal store.

Proof. Let u range over the six directed axial unit vectors. The endpoint window

$$W_u = \{b, b+u, \dots, b+5u\}$$

belongs to $\Omega(b)$. It is dead while b is empty, so among $b+u, \dots, b+5u$ there is an old stone $s_u = b + k_u u$ with $1 \le k_u \le 5$.

Take $z \in B_8(b)$. It lies in a closed $60°$ sector between adjacent directed axes u and v, so

$$z - b = \alpha u + \beta v, \qquad \alpha, \beta \ge 0, \qquad \alpha + \beta = d(b, z) \le 8.$$

After a rotation or reflection, take $u = (1, 0)$ and $v = (0, 1)$, and write $s = s_u = b + ku$ with $1 \le k \le 5$. Then

$$d(z, s) = \max\{|\alpha - k|, \beta, |\alpha + \beta - k|\} \le 8,$$

because α, β, $\alpha + \beta$, and k all lie in $[0, 8]$. Thus $z \in B_8(s) \subseteq \Lambda(P)$. This holds for every z. □

Deadness makes the omitted reply frontier-inert, but it does not impose the same condition on the searched substitute.

Proposition 9.5 (Frontier-certified dead-cell dismissal; companion P1)**.** *Let P be a nonterminal post-opening position with the defender to place. Let $a \ne b$ be legal empty cells. Assume every window in $\Omega(b)$ is already dead in P. Unless $P + a$ is an immediate defender win, also assume*

$$\Lambda(P) \cup B_8(a) = \Lambda(P) \cup B_8(b).$$

Then $b \preceq_n a$ for every finite n. By Lemma 9.4, the displayed condition is equivalent here to $B_8(a) \subseteq \Lambda(P)$. Thus b may be dismissed in favor of any searched a that either wins immediately or is frontier-inert. Deadness of b does not make this separate obligation on a automatic.

Proof idea; complete proof in the companion. Compare $P + a$ and $P + b$ by transposing a and b until the counterpart is used. The hypotheses synchronize legal support. Every window through b is permanently dead, while the remaining changed windows either agree or change ownership in the defender-favouring direction. The occupancy, terminality, and frontier channels therefore give an all-depth causal simulation; an immediate defender win after a is already maximal. See [5]. □

Two hits of one count-4 threat become symmetric only after every other incident window is made inert and the successor supports agree.

Proposition 9.6 (Dead-spoke interchangeable hits; companion P2)**.** *Let P be a nonterminal post-opening position with the defender to place. Let W be an attacker-alive window with exactly four attacker stones and exactly two empty cells $x \ne y$. Both x and y are legal. Assume*

$$(\Omega(x) \cup \Omega(y)) \setminus \{W\}$$

consists entirely of windows already dead in P, and assume

$$\Lambda(P) \cup B_8(x) = \Lambda(P) \cup B_8(y).$$

Then the replies are mutually outcome-dominating at every horizon:

$$x \preceq_n y, \qquad y \preceq_n x \qquad (n < \infty).$$

A solver may retain one canonical member of $\{x, y\}$ at that defender node.

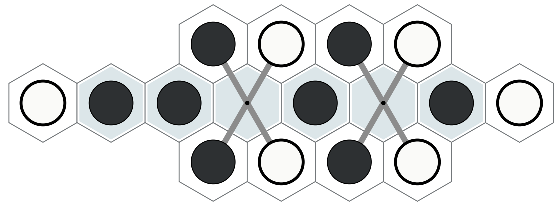


Figure 10: A representative dead-spoke configuration. The highlighted threat has four attacker stones and the two marked empty hits. Three spoke directions per hit show old attacker and defender witnesses. The P2 premise applies this deadness condition to all 17 other windows through each marked cell, not only the representative spokes shown.

Proof idea; complete proof in the companion. Either defender hit kills W. The dead-spoke premise then makes every window through either special cell permanently dead. Transposing x and y preserves occupancy choices, the support equality preserves legal moves in both directions, and every mask outside the dead union is identical. This is an outcome-preserving bisimulation at all depths. See [5]. □

Equality of count profiles is insufficient. Labelled empty cells and the window-intersection hypergraph can change minimum hitting sets or fork-creating cells even when every pair $(\mathrm{cnt}_A, \mathrm{cnt}_D)$ agrees. A broader equivalence would have to preserve labelled masks, incidence, occupancy, and radius-eight support throughout the n-causal cone. The dead-spoke premise avoids that unproved classification.

9.3 Same-turn commutation

Two placements may be sorted only when both were legal before the turn and neither possible first placement wins.

Proposition 9.7 (Nonwinning-prefix turn commutation; companion P3)**.** *Let P be a nonterminal* FirstStone *position for either player X. Let $p \neq q$ be legal already in P, and suppose neither $P + p$ nor $P + q$ is terminal. Then both traces $p; q$ and $q; p$ are legal and have the same qualitative rule outcome after the second placement. If nonterminal, they have identical occupancy, labelled window masks, legal support, player to move, and* FirstStone *phase. If terminal, both terminate on the second placement with winner X and the same absolute placement count.*

Proof idea; complete proof in the companion. Initial legality makes either second cell remain legal, and the final support $\Lambda(P) \cup B_8(p) \cup B_8(q)$ is order-independent. Inserting the two owner bits into every window mask commutes, including in a window containing both cells. The singleton-nonterminal premises exclude a first-placement win; any joint win is therefore completed by the second placement in both orders. See [5]. □

The live engine states need not be field-identical. Ordered history, last-turn metadata, serialized order, and the retained terminal SecondStone first-cell payload may differ. The proposition equates rule outcomes, not these order-sensitive features or the two intermediate SecondStone threat states.

The analytic proofs have randomized local checks, not exhaustive machine proofs. MV-P3 tested 2,997 random legal same-mover pairs under the singleton-nonwin filter and found 0 commutation mismatches. MV-L7 tested 400 adversarial dead-cell configurations and found 0 legality-frontier violations; the worst minimum distance from a cell of the radius-eight ball to an old stone was 6, leaving margin 2. Full exhaustive quotient enumeration remains open.

$$
\begin{tikzcd}
P \arrow[r, "p"] \arrow[d, "q"'] & P+p \arrow[d, "q"] \\
P+q \arrow[r, "p"'] & P+\{p,q\}
\end{tikzcd}
$$

Figure 11: The commuting square for two same-turn placements. Both paths reach the same completed-turn rule state under the premises of P3.

These results do not claim that an arbitrary frontier-extending reply can substitute for a dead cell, that equal window counts make hitting cells interchangeable, or that stones of one colour are globally monotone-helpful when legal frontiers differ; no transferred strategy continues after a terminal placement, and the prospective exhaustive enumerations are specifications rather than executed evidence.

10 Mechanized validation

The validation harness uses a mini-model solver that was checked move-for-move against the *hexo_engine* implementation. The solver machine-checks the finite claims used by the closure experiments.

Two closure experiments were executed. First, the closure-restricted solver did not claim the engineered junction position. The junction cell was in the closure, and the defender survived after 1,396 uncapped nodes. Second, a bounded random check covered 91 attackable positions. The closure-restricted solver made 52 WIN claims, with no divergence from the full legal move set at the matched horizon; the run took 139 seconds. As a negative control, the earlier radius-2 experiments used the same harness and produced false WIN claims for unsound restrictions.

These experiments are consistency checks between the implemented closure and Theorem 6.6. They are not part of the proofs of Theorems 5.1, 6.1 and 6.6. The separate machine-verification status of the domination patterns is summarized in Section 9.

11 Open problems

Sharpened budget. The completion guard in Theorem 6.1 counts every defender placement before the horizon. A sharper budget would count quiet placements F together with a per-window forced-hit capacity H_W. Proving this bound requires per-branch worst-case bookkeeping over the certificate DAG. The soundness result in Theorem 5.1 does not require the sharpening, at the cost of wider zones for deep loose certificates.

Sharper frontier bands. The frontier guard Item (Z5) uses the worst-case radius $8D_N$ needed by the chain argument in Lemma 4.10. Each link also costs defender tempo, and the completion guard already limits that tempo. A joint accounting of time and distance should shrink the band substantially; that accounting is unwritten.

Pairing at the threshold. The density bound in Corollary 7.2 leaves the case $k = 7$ and $g = 3$ open, where the required matched density is exactly one. This threshold case does not affect Hexo, whose winning length is $k = 6$, and Theorem 7.1 already rules out the required Hexo pairing. Existence at the threshold remains unsettled.

Formalization. The model and results through Theorem 7.1 are finite combinatorics and are feasible to formalize in Lean or Coq. Such a formalization would yield a verified reference checker for Items (Z1), (Z2), (Z4) and (Z5).

A Notation reference

Table 1: Notation used in the main text.

Symbol	Meaning	Defining location
$d(x, y)$	Hex distance between cells x and y.	Definition 2.1
$\mathfrak{a}$	An axis vector in $\{(1, 0), (0, 1), (1, -1)\}$.	Definition 2.2
W	A six-cell window along one axis.	Definition 2.2
$\mathrm{cnt}_X(W, P)$	Number of stones of player X in W at P.	Definition 2.4
alive, dead, threat	Window states determined by the two colour counts.	Definition 2.6
own_win_now(P)	Current-turn completion predicate for the mover.	Definition 2.7
$\mathrm{hit}(P)$	Multiset of empty-cell sets of opponent threat windows.	Definition 2.7
$\mathrm{mhs}(P)$	Minimum size of a set meeting every member of $\mathrm{hit}(P)$.	Definition 2.7
ply	One placement, indexed absolutely along a play.	Definition 2.8
$\mathfrak{D}(P, T)$	Defender placements strictly before horizon ply T.	Definition 2.8
$\mathrm{core}(\mathcal{C}, N)$	Certificate cells and witness windows below node N.	Definition 4.7
$\mathrm{Prot}(N)$	Core together with defender completion-guard empties.	Definition 4.8
$S(N)$	Searched legal defender replies at AND node N.	Definition 4.6
$\mathrm{Legal}(P)$	Legal empty placement cells in position P.	Definition 2.5
$\mathrm{Stones}(P)$	Occupied cells in position P.	Definition 2.4
$B_r(U)$	Cells within distance r of some cell of U; written $B(r; U)$ in the main theorem.	Definition 4.8
$\mathrm{hitting}(P)$	Union of the empty sets of attacker-threat windows.	Theorem 6.1
$R(P, D)$	Horizon relevance-closure candidate generator.	Definition 6.5
X	Real-only dismissed defender stones in the coupling.	Definition 4.11
Y	Ghost-only searched filler defender stones in the coupling.	Definition 4.11
$\widehat{X}$	History of every cell ever added to X.	Definition 4.11
$\Phi(P)$	Dynamic potential of attacker-touched alive windows.	Definition 8.1
λ	Potential base $\sqrt{3}$.	Definition 8.1
$\Psi_F(P)$	Residual potential for a fixed target family F.	Definition 8.1

Table 2: Crosswalk from paper numbering to source tags in the repository proof document.

Paper number	Source tag	Name
Definition 2.1	D1	Board, cells, and distance
Definition 2.2	D2	Axes and windows
Definition 2.4	D3	Positions
Definition 2.5	D4	Legality and transitions
Definition 2.6	D5	Window states
Definition 2.7	D6	λ^1 predicates
Definition 2.8	D7	Plies and horizon
Definition 2.9	D8	Dominance direction
Definition 4.6	D9	Certificate grammar
Definition 4.7	D10	Certificate core
Definition 4.8	D11	Zone-carrying certificate
Definition 4.11	D12	Real–ghost coupling
Definition 6.5	D13	Horizon relevance closure
Fact 2.3	F1	Window collinearity
Lemma 3.1	L1	Packing bound
Lemma 3.2	L2	Boundary pair
Lemma 4.1	L3	Difference accounting
Lemma 4.2	L4	Permanence
Lemma 4.3	L5	Inert-extension transfer
Lemma 4.4	L6	Deletion monotonicity and well-formed legality
Lemma 4.5	L7	Completion counting
Proposition 4.9	(unnumbered)	Monotonicity of protection
Lemma 4.10	L9	Frontier chain closure
Lemma 6.2	L10	Short-range attacker-core containment
Theorem 3.3	T1	Single-turn tactical locality
Theorem 3.4	T2	Threat-creation locality
Theorem 5.1	T3	Horizon-parameterized dismissal soundness
Theorem 6.1	T4	Zone reading
Theorem 6.3	T5	Static-set coverage for short horizons
Theorem 6.4	T6	Forced-tree regime
Theorem 6.6	T7	Closure and certificate-extras coverage
Theorem 7.1	T8	No pairing
Corollary 7.2	T8.1	Translation-invariant line grids
Item (Z1)	Z1	Hitting condition
Item (Z2)	Z2	Protection condition
Item (Z4)	Z4	Well-formed attacker legality
Item (Z5)	Z5	Horizon-scaled frontier guard
Equation (MI)	MI	Anchored defender-count inequality

Acknowledgments

The underlying proof documents were drafted and adversarially reviewed with the assistance of AI systems (Claude and GPT-family models) and machine-checked where stated.

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